

Mayushan Mayurathan

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EDUCATION

University of Toronto

Sep 2021 – June 2026

Honours Bachelor of Science in Economics & Computer Science

- **Relevant Computer Coursework:** Database Management Systems, Data Structures & Algorithms, Operating Systems, Systems Programming, Discrete Mathematics
- **Relevant Economics/Statistics Coursework:** Capital Markets & Asset Pricing, Corporate Finance, Exploratory Data Analysis, Applied Econometrics, Algorithms of Data Analysis

EXPERIENCE

Software Systems Analyst

Sept 2024 - Dec 2024

Royal Bank of Canada (RBC)

Toronto, ON

- Engineered **data extraction pipelines** by integrating **SQL queries** with **Python** using the **pyodbc library**
- **Automated CPU Utilization Reporting** through a combination of **Ansible and python scripting**
- Administered Unix scripts for Oracle database maintenance to minimize discrepancies across server environments
- Maintained over **300+ Oracle and Sybase system devices** by regulating **ServiceNow** compliance

Quantitative Analyst

May 2024 – Aug 2024

Northstar Trading

Toronto, ON

- Developed a high-performance predictive model for power markets (**88% R^2**), applying **regression and machine learning**; **outperformed existing benchmarks by 4.5% in forecast accuracy**
- Built **low-latency Python ETL pipelines** to transform and stream data to traders giving access to actionable signals
- Designed **statistical arbitrage** strategies for congestion spreads using **non-parametric tests** and **time-series analysis**
- Modeled ISO grid topology as a graph structure using PandaPower and MATLAB; explored graph neural networks in TensorFlow to capture network-level dynamics and forecast disruptions

Mobile App Developer

Jan 2024 – May 2024

Fibra Inc.

Toronto, ON

- Developed Bluetooth support and data storage app features using **React and JavaScript** for iOS/Android compatibility
- Engineered and designed Fibra Inc's **Firebase database** to improve data quality and **cut redundancy by 80%**
- Engineered seamless and reliable **Wifi and Bluetooth data paths** achieving **99% uptime**

Software Engineer

Feb. 2023 – May 2023

Project: Human City

Toronto, ON

- Developed a reputation ranking system for clients and vendors using Python to create user credibility in the application
- Utilized **Figma** to create interactive prototypes and detailed UI mock-ups, streamlining design feedback cycles by 30%
- Presented live technical walkthroughs to the team to bolster understanding of algorithm structure and implementation

PROJECTS

Boombox.Ai | [JavaScript](#), [HTML](#), [CSS](#), [Python](#), [React](#), [Axios](#), [OpenCV](#), [DeepFace](#), [Cohere NLP](#), [RESTful APIs](#)

- Led a team of 4 in the development of Boombox.ai, a interactive web app for the UofTHacks11 hackathon using **React**, **JavaScript**, **HTML**, **CSS**, and integrated Spotify API using **Axios** for music streaming functionalities.
- Integrated **AI technologies** using **Cohere NLP** for sentiment analysis and **DeepFace with OpenCV** (in Python) for emotion recognition, mapping user emotions to curated playlists of nostalgic songs.

UofT Course Timetable System | [Java](#), [JavaFX](#), [ControlsFX](#), [RESTful API](#)

- Led a team of 8 developers to develop a timetable generator in **Java** adhering to SOLID design principles, Clean Architecture, and test-driven development to maintain testing with **95%+ code coverage**
- Queried the UofT and Rate My Prof REST APIs to generate **100 thousand+** timetables in under 3 seconds and created a tagger and filtering system to filter timetables and narrow search results by 70%+

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS, C/C++, R, VBA, MIPS Assembly, Bash, LaTeX

Technologies: React, AWS, Axios, Firebase, Postman, JSON, STATA, JavaFX, ControlsFX, JUnit, Cohere

Developer Tools: Git, Figma, Expo, Miro, Linux, Jupyter, Collab, Power BI

Libraries: LangChain, Pandas, GeoPandas, NumPy, SciPy, Matplotlib, Beautiful Soup, OpenPyXL, DeepFace, OpenCV